

2026 | Session – 1 COMP2750/6750: APPLICATIONS MODELLING AND DEVELOPMENT

Assessment Task 3: Application Development and Testing

Submission details	
Due Date	<i>Submission Due Week-13 in your registered SGTA's</i>
Weighting	<i>This assessment will be marked out of a total of 100 marks. This assessment will contribute to 30% of overall unit grade</i>
Time to complete	<i>This assessment will take approximately 30 hours (individual hours) to complete</i>
Length & Format	<i>PPT/PDF, Codes. See task details for specific requirements per section.</i>
Late Penalties	<i>Standard late penalty applies. Failure to present in your registered SGTA will result in a mark of zero for the presentation component unless special consideration is approved.</i>
How to Submit	<i>Two submissions required:</i> <i>(1) Group submits project files plus the presentation as a zip file via iLearn assignment link before your SGTA. Group presents the application live during the SGTA.</i> <i>(2) Individual submits Figma snapshots plus the test case document as a PDF via a separate iLearn link before your SGTA.</i>
Return of Assessment Grade & Feedback	<i>Grades and detailed feedback will be returned via iLearn within 14 working days of submission.</i>
Purpose	<i>In Assessment 1, you analysed the Campus Marketplace case study as a business analyst, identifying stakeholders, writing requirements, and modelling the system. In this assessment, you bring that system to life. Working in groups, you will design and build a working web application for a simplified version of Campus Marketplace using HTML, CSS (Bootstrap), JavaScript, and Firebase. You will then individually design your dream screen for the case study and write test cases for the application demonstrating your understanding of software testing.</i> <i>This assessment mirrors authentic development work: translating requirements into a working product, integrating with a cloud database for persistent storage, collaborating as a team, and verifying that the application works through structured testing.</i>
Learning Outcomes Assessed	<i>ULO1:</i> <i>Demonstrate ability to ethically communicate software requirements and designs, clearly and effectively</i> <i>ULO2:</i> <i>Practice the key phases of the SDLC including requirements engineering, analysis, design, basic development and testing</i> <i>ULO3:</i> <i>Demonstrate understanding of alternative SDLC lifecycle models</i> <i>ULO4:</i> <i>Demonstrate an understanding of the concepts and tools needed to successfully design and build an application</i> <i>ULO5:</i> <i>Integrate an application with a database or other form of persistent storage</i>
Skills assessed	<i>This task allows you to demonstrate:</i> <i>(1) Your ability to design a web application using Figma</i> <i>(2) Your ability to build a functioning web application using HTML, CSS, JavaScript, and Firebase</i>

	(3) Your capability to integrate a frontend with a cloud database (Firestore) for persistent data storage (4) Your understanding of user authentication and data management (5) Your ability to present and explain technical work clearly to an audience (6) Your understanding of software testing through well-structured test cases
How will this task be assessed?	Group (70%) and Individual (30%); Group work is assessed via live presentation and demo during the SGTA. Individual work is assessed via Figma design (which should be included in the presentation), and a test case document with two test cases.
Use of AI	Open. You may use AI to help brainstorm ideas, debug code, or generate examples. However, your submitted application must be your own group's work, and you must be able to explain every part of your code during the presentation. Individual test cases must be your own work.
Short automated extensions	No. This assessment is presented in your registered SGTA in Week 13. Extensions are not available.

Task overview

Your group will build a simplified version of the Campus Marketplace web application introduced in the A1 case study. The application allows Macquarie University students to sign in, browse items listed by other students, view their own listings, and shortlist items they are interested in.

This is not a full implementation of everything described in the A1 case study. You are building a focused, working subset that demonstrates the core concepts of application development and database integration.

Additionally, you will also individually design a "dream" screen of the Campus Marketplace in Figma and write test cases that are presented and executed live.

Task details / specific instructions

Part A: Group Application Development and Presentation (70 marks)

Your group will build a web application using HTML, CSS (Bootstrap), JavaScript, and Firebase (Authentication and Firestore), and present it live during your Week 13 SGTA.

Page 1: Sign-In Page (login.html)

A sign-in page where users enter their email and password. Firebase Authentication verifies the credentials. On successful sign-in, the user is redirected to the Welcome page. Users who are not signed in cannot access any other page in the application.

You do not need to build a sign-up page. Create your test user accounts directly in the Firebase Console.

Page 2: Welcome Page (index.html)

This is the landing page after signing in. It welcomes the user by displaying their email and provides navigation links to the three main areas of the application:

- (1) Browse Marketplace - links to marketplace.html
- (2) My Listings - links to mylistings.html
- (3) My Shortlist - links to shortlist.html

A Sign Out button is also present on this page. Clicking it signs the user out of Firebase and redirects them back to login.html. If a user navigates to this page without being signed in, they are automatically redirected to login.html.

Page 3: Marketplace (marketplace.html)

Displays all items listed by other users. Items where the seller is the currently signed-in user must not appear here. Each item displays at minimum: the item name, a description, the price (or "Trade" if the item is for trading), the category (e.g., Textbooks, Furniture, Clothing, Electronics), an image, and the seller's email.

A "Shortlist" button on each item allows the user to save it to their shortlist. If the user has already shortlisted an item, the button should indicate this (e.g., show "Shortlisted" in a different colour and be disabled). This page includes a link back to the Welcome page and a Sign Out button. If a user navigates here without being signed in, they are automatically redirected to login.html.

Page 4: My Listings (mylistings.html)

Displays only the items listed by the currently signed-in user. Users can see what they have listed but cannot shortlist their own items. This page allows the user to view all items they have added to the items collection in Firestore where the seller ID matches their own user ID. This page includes a link back to the Welcome page and a Sign Out button. If a user navigates here without being signed in, they are automatically redirected to login.html.

Page 5: My Shortlist (shortlist.html)

Displays items the signed-in user has shortlisted from the Marketplace page. Each shortlisted item shows the item name and a "Remove" button. Shortlisted items are stored in a Firestore collection linked to the user's ID and the item's ID. Users can remove items from their shortlist, which deletes the corresponding document from Firestore. This page includes a link back to the Welcome page and a Sign Out button. If a user navigates here without being signed in, they are automatically redirected to login.html.

Data Setup:

As a group you will need to add your data in the Firebase Console (the same way you added events during the workshops). Images are placed in the project's images folder. The group must **have at least 8 items total across at least 3 categories, spread across at least 3 test user accounts** so that the Available Items and My Listings filtering works correctly during the presentation. Create at least 3 test user accounts in the Firebase Console.

Group Deliverables: Submit as one/Present the demo as a team

Project files submitted as a zip file via iLearn before your SGTA. The zip must include all HTML files, JavaScript files, CSS files (if any), the images folder, and a README.txt file containing your Firebase configuration details and test user credentials (email and password) so the marker can test your application, if need be.

(1) **Presentation slides** (PowerPoint) included in the zip file.

Live presentation and demo during your Week 13 SGTA.

Live Presentation (10 minutes):

Your group presentation must be **no longer than 10 minutes**. Keep it focused. The presentation must include:

A live demonstration of the working application showing signing in, browsing available items loaded from Firestore, viewing My Listings (showing only the signed-in user's items), shortlisting an item, viewing

the shortlist, removing a shortlisted item, signing out and signing in as a different user to show different available items and a separate shortlist (3-5 minutes)

A **brief code walkthrough (when probed)** explains one section of the code they worked on, demonstrating understanding of how the frontend connects to Firebase (2-3 minutes).

Every teammate presents their Figma design (explained in Part-B) (2 minutes).

Every group member must speak during the presentation. A group member who does not present or cannot explain their contribution may receive a reduced individual mark.

Part B: Individual Figma Screen Design (10 marks)

Submitted individually as a PDF export via iLearn before your SGTA.

The application you built in Part A is functional but basic, limited by the constraints of what you learned in the workshops. For this task, imagine you had no technical constraints. If you could design the perfect Campus Marketplace screen, what would it look like?

Using Figma (figma.com, free for students), design one screen (a website) of the Campus Marketplace application. This can be any screen, the main marketplace browse page, a detailed item view, a user profile page, a messaging screen between buyer and seller, a transaction meetup screen, a sustainability dashboard, or anything from the A1 case study.

Your design must:

Be a **high-fidelity mock-up**: it should look like a real, polished application, not a wireframe or sketch

(1) Include **realistic content** (real item names, descriptions, prices, images, not placeholder "Lorem ipsum" text)

(2) Demonstrate **thoughtful user experience choices**: layout, visual hierarchy, navigation, and how the user would interact with the screen

Be clearly **labelled with a title** indicating which screen it represents.

Part C: Individual Test Cases with Live Walkthrough (20 marks)

Test case document submitted individually as a PDF via iLearn before your SGTA.

Each student will write two test cases for the Campus Marketplace application built in Part A. **Your test cases will be executed against your group's application**

Each test case must include:

Component	Description
Test Case ID	A unique identifier (e.g., TC-01)
Test Case Title	A brief descriptive name
Description	What this test case is verifying
Preconditions	What must be true before the test can run
Test Steps	Numbered step-by-step instructions that someone else could follow to execute the test
Expected Result	What should happen if the application is working correctly
Actual Result	What actually happened when you ran the test (fill this in after testing)
Pass/Fail	Did the actual result match the expected result?

Quality Criteria

A high-quality submission:

- Has a working application that runs without errors when opened with Live Server
- Loads all data dynamically from Firestore, no hardcoded items in the HTML
- Correctly separates Available Items (other users' items) from My Listings (your own items)

- Shows per-user shortlist functionality with persistence across sessions
- Includes a concise, well-rehearsed group presentation where ideally every member speaks
- Features a polished Figma screen design that demonstrates creative thinking beyond the coded application
- Contains two well-structured test cases that have been executed against the real application
- Demonstrates confident, clear live execution of test cases during the individual walkthrough
- Has clean, readable, commented code
- Includes a README.txt with Firebase config and test user credentials

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The rubric below guides you to what we expect.

Part A: Group Application and Presentation (70 marks)

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Sign-In and Authentication (12 marks)	Sign-in works flawlessly. Unauthenticated users redirected to login. Sign-out works and redirects.	Sign-in and sign-out work correctly. Authentication checks present. Minor issues.	Sign-in works. Authentication check may be inconsistent. Sign-out present.	Sign-in partially works or has bugs. Limited authentication checking.	Sign-in does not work or is not connected to Firebase.
Available Items from Firestore (12 marks)	Items load dynamically from Firestore. Correctly excludes signed-in user's own items. All required fields displayed clearly. Layout is polished and responsive.	Items load from Firestore with correct filtering. All required fields present. Minor display issues.	Items load from Firestore. Filtering mostly works. Some fields may be missing or poorly formatted.	Items partially load or filtering is incorrect. Limited Firestore integration.	Items hardcoded in HTML or do not load from Firestore. No user-based filtering.
My Listings (12 marks)	Correctly shows only the signed-in user's items. All fields displayed. Clear visual separation from Available Items. Different users see different listings.	Shows user's own items correctly. Most fields present. Minor issues.	My Listings section exists and mostly shows correct items. Some display issues.	Section exists but filtering is incorrect or incomplete.	No My Listings section or not connected to Firestore.

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Shortlist Functionality (12 marks)	Shortlist saves to Firestore per user. My Shortlist displays correctly. Remove works. Different users see different shortlists. Cannot shortlist own items. Duplicate prevention present.	Shortlist saves and displays per user. Remove works. Minor issues.	Shortlist saves to Firestore. Display or remove may have bugs. Per-user separation works.	Shortlist partially works. May not be per-user or may not persist.	No shortlist functionality or not connected to Firestore.
Sign Out and Access Control (5 marks)	Sign out works cleanly. Redirects to login. Cannot access marketplace without signing in. Session handled correctly.	Sign out and redirect work. Access control present. Minor issues.	Sign out works. Access control may be inconsistent.	Sign out partially works or no redirect.	No sign out or no access control.
Presentation: Demo and Delivery (5 marks)	Concise, well-rehearsed demo. All features demonstrated smoothly. Multiple users shown.	Good demo covering all features. Minor hiccups.	Demo covers most features. Most members speak. May need restarts or prompts.	Demo incomplete or disorganised. Some members do not speak. Significant issues.	Presentation not delivered or application does not run.
Code Quality and Explanation (12 marks)	Clean, commented, readable code. Each member explains their section with clear understanding of frontend-backend	Good code structure. Members explain their sections with reasonable understanding.	Code works but is messy or uncommented. Basic understanding shown when explaining.	Code is disorganised. Members struggle to explain their contributions.	Code does not work or members cannot explain any part of it.

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
	connection. Can answer follow-up questions.				

Part B: Individual Figma Screen Design in Live Walkthrough (10 marks)

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Visual Design Quality (10 marks)	Polished, professional high-fidelity design. Consistent typography, colour scheme, and spacing. Looks like a real production application. Creative and thoughtful.	Clean, well-designed screen. Good use of colour, typography, and layout. Minor inconsistencies.	Reasonable design effort. Mostly consistent styling. Some elements feel unfinished.	Basic design. Looks like a wireframe with colour. Minimal polish.	No meaningful design effort. Placeholder content. Sketch-level quality or not submitted.

Part C: Individual Test Cases (20 marks)

Assessment Component	Outstanding High Distinction (100%-85%)	Advanced Distinction (84%-75%)	Proficient Credit (74%-65%)	Functional Pass (64%-50%)	Developing Fail (49%-0%)
Test Case –1 Quality (10 marks)	Every test case has clear, specific, numbered steps someone else could follow. Expected results are precise and	Well written with clear steps and expected results. Minor ambiguities. Pass/fail mostly accurate.	Present but steps may be vague or expected results imprecise. Some actual results recorded	Lack detail. Steps unclear or missing. Expected results vague. Limited evidence of	Incomplete or demonstrate no understanding of testing. No evidence of test execution.

	verifiable. Preconditions accurate. Actual results documented accurately. Pass/fail correctly determined. Failures explained honestly.		but evidence of execution inconsistent.	execution. Actual results mostly empty or generic.	
Test Case –2 Quality (10 marks)	Every test case has clear, specific, numbered steps someone else could follow. Expected results are precise and verifiable. Preconditions accurate. Actual results documented accurately. Pass/fail correctly determined. Failures explained honestly.	Well written with clear steps and expected results. Minor ambiguities. Pass/fail mostly accurate.	Present but steps may be vague or expected results imprecise. Some actual results recorded but evidence of execution inconsistent.	Lack detail. Steps unclear or missing. Expected results vague. Limited evidence of execution. Actual results mostly empty or generic.	Incomplete or demonstrate no understanding of testing. No evidence of test execution.